

Total No. of Questions : 8]

[Total No. of Printed Pages : 3

Roll No

CE-5003-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

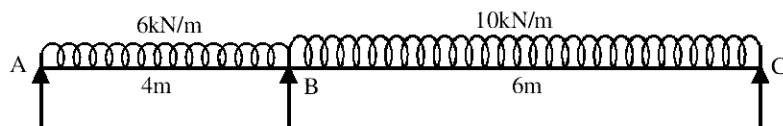
Structural Analysis - II

Time : Three Hours

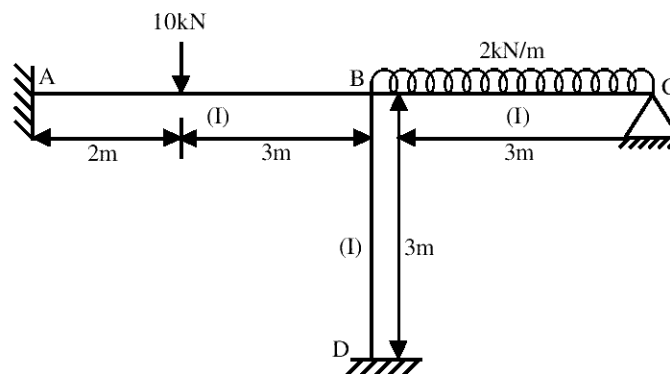
Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Be precise in your answers.

1. Analyse the continuous beam by moment distribution method: 14



2. Analyse the beam by Kani's method. 14



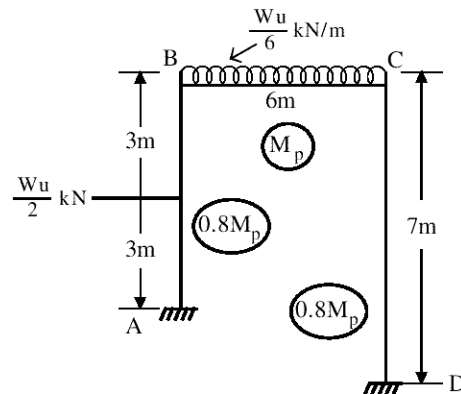
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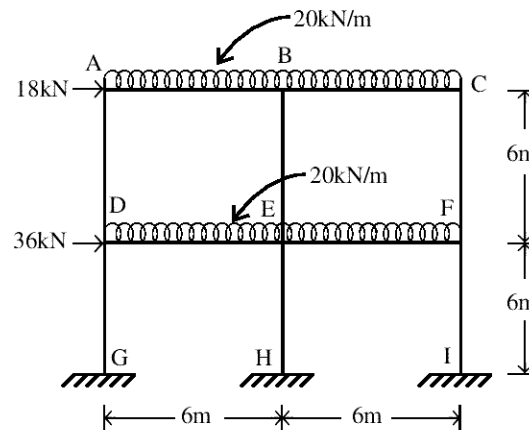
3. Find the collapse load for the loaded frame. The plastic moment capacity for beam is M_p and for column $0.8M_p$ respectively is given below

14



4. Analyze the building frame using by portal method. Assume point of inflection at 0.11 for gravity loads. Sketch BMD (figure)

14

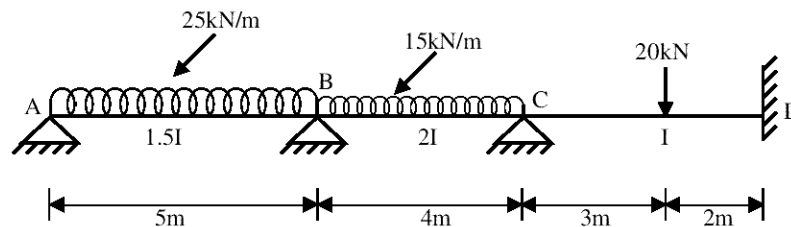


5. Explain influence line diagram for intermediate structures and the Muller Breslau principle.

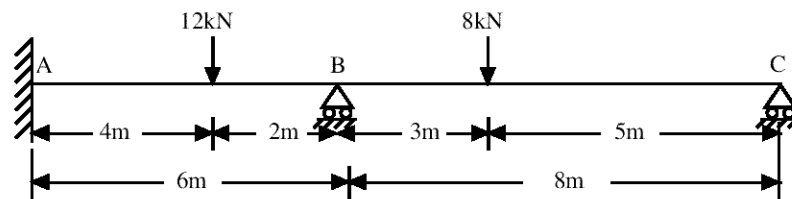
14

[3]

6. Analyze the continuous beam shown in figure by stiffness method. Draw bending moment diagram and elastic curve. 14



7. Analyse the continuous beam as shown in figure. By Flexibility matrix method. 14



8. Define the following: 14
- i) Distribution factor
 - ii) Stiffness coefficients
 - iii) Factor of safety
 - iv) Working load
 - v) Plastic moment capacity
 - vi) Load factor
